

NIHAL POTHUNOORI

804-549-9190 | nihal4@illinois.edu | <https://nihalpothunoori.github.io/> | US Citizen

EDUCATION

University of Illinois at Urbana-Champaign

Expected May 2027

B.S. in Electrical Engineering

Relevant Coursework: Control Systems; Principles of Safe Autonomy; Analog Signal Processing; Computer Systems & Programming

SKILLS

Programming: Python (PyTorch, OpenCV, NumPy), C/C++, Matlab, Java, C#, .NET

Robotics: ROS2, Simulation (Gazebo, Isaac Sim/Lab), VLAs, Deep/Reinforcement Learning, Diffusion, State Estimation, Path Planning

Design: CAD (Fusion360), Altium, KiCAD, PCB Design (I2C, SPI)

EXPERIENCE

DASLABS, University of Illinois

Jan 2026 - Present

Undergraduate Researcher

Champaign, IL

- Leveraging diffusion-based path planning to generate high-fidelity backbone trajectories for a hybrid (rigid + soft) arm; simulating possible trajectories and sampling from a high-reward distribution; implemented domain randomization for real-world accuracy.
- Implementing vision-language-action (VLA) models to identify stable anchor locations for soft robots, enabling complex locomotion and manipulation tasks within highly constrained or confined environments.

College of Education, University of Illinois

Aug 2025 – Present

Web Developer/IT Support

Champaign, IL

- Built an automated enterprise-level permissions management system for ~2,500 users via a high-performance .NET 8 C# library integrated with Azure Active Directory to streamline identity and access management across the department.
- Developing an intuitive web-based interface in HTML to enable non-technical administrators to execute complex operations on user roles, reducing manual IT support tickets and ensuring 99.9% consistency in organizational permission hierarchies.

Autonomous Materials Systems Group, University of Illinois

Aug 2025 – Jan 2026

Undergraduate Researcher

Champaign, IL

- Designed a 3-axis robotic manipulator using inverse kinematics and automated pipette dispenser for the preparation of DCPD polymer samples within a DARPA-funded laboratory focused on rapid, autonomous material synthesis and chemical discovery.
- Architected and deployed a PyTorch-based reinforcement learning framework for closed-loop process control, optimizing between environmental variables and mechanical actuation to achieve a 75% reduction in total sample preparation time.

Toast Gastropub

June 2025 – August 2025

Line Cook

Richmond, VA

PROJECTS

Rubik's Cube Solver

Jan 2026 – Present

- Utilizing dual robotic arms (OMX) to solve a Rubik's Cube in real-time with a Vision-Language-Action (VLA) pipeline that perceives the six-sided cube state from multi-view camera feeds; closed-loop state estimation pipeline to resolve unknown cube states.
- Training robotic arms via Inverse Reinforcement Learning (IRL) to master foundational manipulation policies, such as 90-degree face rotations and orientation-aware re-grasping; implementing action-chunking for more fluid motion.

Laundry Robot

Dec 2025

- Developed a ROS2 mobile platform to autonomously retrieve scattered laundry; uses a YOLOv8 segmentation model for clothing detection; perception utilizes a vision-to-world service to map 2D image pixels into 3D coordinates in lieu of an RGB-depth camera
- Implemented an Extended Kalman Filter (EKF) to fuse asynchronous odometry and visual data for localization and state estimation; estimates informed a Stanley Controller that minimizes cross-track error to provide precise path-following controls

Wearable EMG Instrument (2nd Place - International IEEE Arduino Contest)

Aug 2024

- Designed high-gain op-amp circuitry with active filter stages to capture EMG impulses from biceps while suppressing interference.
- Engineered a wearable instrument that maps muscle contraction intensity to musical notes to enable gesture-based performance.

LEADERSHIP

Hardware Lead

Jan - May 2025

iRobotics Micromouse, University of Illinois

Champaign, IL

- Led hardware design for maze-solving robots; ran weekly design reviews and taught PCB/PCBA fundamentals (15+ members).
- Fabricated custom PCBs in KiCAD with I2C/SPI sensor interfaces and analog filtering to ensure signal integrity for data acquisition.